COMBINATORICS ON ONE PAGE

Pascal's triangle: coin tosses and multiples of 5 and 7

Row n counts the ways to get k heads in n tosses, and reading its entries modulo a prime leaves triangles of multiples.

1 / THE RULE AND THE PICTURE

Rows are coin-tossing counts, and the multiples of 5 among them follow a pattern

											row sum
					1						$2^0 = 1$
				1		1					$2^1 = 2$
			1		2		1				$2^2 = 4$
		1		3		3		1			$2^3 = 8$
	1		4		6		4		1		$2^4 = 16$
	1		5		10		10		5	1	$2^5 = 32$
1		6		15		20		15		6	1	$2^6 = 64$
	1		7		21		35		35		21		7		1	$2^7 = 128$
	1		8		28		56		70		56		28		8		1	$2^8 = 256$
1		9		36		84		126		126		84		36		9		1	$2^9 = 512$
1	10		45		120		210		252		210		120		45		10	1	$2^{10} = 1024$

ten coins, five heads: $\frac{252}{1024} = \frac{63}{256}$

Each entry is the sum of the two above it, and the entry $\binom{n}{k}$ counts the strings of n tosses with exactly k heads, so a row adds up to 2^n:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}, \quad \sum_k \binom{n}{k} = 2^n$$

With fair coins the chance of k heads is $\binom{n}{k}/2^n$. In red are the multiples of 5: none in rows 0 to 4, four in row 5, and in row 10 only the ends and 252 are left.

2 / BINARY STRINGS

Four tosses sorted by the number of heads

heads	the 16 strings of 4 tosses	count	chance
0	TTTT	1	$\frac{1}{16}$
1	HTTT THTT TTHT TTTH	4	$\frac{4}{16}$
2	HHTT HTHT HTTH THHT THTH TTHH	6	$\frac{6}{16}$
3	HHHT HHTH HTHH THHH	4	$\frac{4}{16}$
4	HHHH	1	$\frac{1}{16}$

Exactly half heads has probability $\frac{1}{2}$ for 2 tosses, 0.375 for 4 and $\frac{252}{1024} = 0.246$ for 10.

3 / LUCAS'S THEOREM

The residue splits digit by digit

For a prime p, with $n = \sum n_i p^i$ and $k = \sum k_i p^i$ in base p,

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}$$

so $\binom{n}{k}$ is a multiple of p exactly when some digit k_i exceeds n_i, and row n has $\prod_i (n_i + 1)$ entries that are not multiples. Rows 0 to $p-1$ have none. For $n = 10 = 20_5$ and $k = 5 = 10_5$ the product is $\binom{2}{1}\binom{0}{0} = 2$, which is $252 \bmod 5$.

4 / COUNTING THE SURVIVORS

The entries of a row that are not multiples of p number $\prod(n_i + 1)$

p	row n	n in base p	entries not divisible by p	note
5	10	20_5	3	$1,\ 252,\ 1 \bmod 5$ gives $1, 2, 1$
5	24	44_5	25	every entry of the row
5	25	100_5	2	only the two ends
7	14	20_7	3	$1,\ 3432 \equiv 2,\ 1$
7	48	66_7	49	every entry of the row

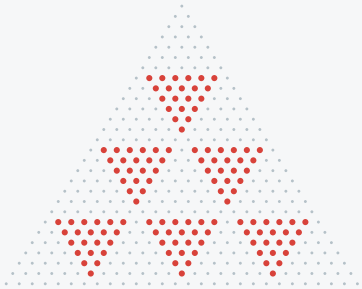
A row all of whose entries survive is row $p^m - 1$, whose digits are all $p - 1$, and a row with only two survivors is row p^m.

5 / THE MULTIPLES AS TRIANGLES

Inverted triangles of $\frac{(p-1)p}{2}$ multiples, j of them in the jth block of p rows



multiples of 5, rows 0 to 24



multiples of 7, rows 0 to 27

In each picture the red dots are the multiples of p and the grey dots the rest. The first block of p rows, from row p, holds one inverted triangle with sides $p - 1$, which has 10 dots for $p = 5$ and 21 for $p = 7$, the next block holds two, then three, and in rows 0 to 24 there are 10 triangles, 100 multiples of 5 among 325 entries. At scale p^2 the whole pattern repeats: the first p^2 rows contain $\left(\frac{p(p+1)}{2}\right)^2$ entries that are not multiples, 225 for 5 and 784 for 7.

ASIDE / TRY THESE

Why is $\binom{25}{5} = 53130$ a multiple of 5? In base 5, $25 = 100_5$ and $5 = 10_5$, so $k_1 = 1 > n_1 = 0$. For $p = 2$ the same rule leaves the Sierpinski triangle.

LINKS TO OTHER SHEETS

The partitions sheet counts the same bit strings; the fractions tree sheet builds its rows from the same binary branching.

Convention. $\binom{n}{k}$ the number of ways to choose k of n; n_i, k_i the base-p digits; all residues, counts and Lucas's theorem were checked by computation.

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